Md Masudur Rahman

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Residential Status: US Permanent Resident

RESEARCH INTERESTS

Artificial Intelligence and Machine Learning: Reinforcement Learning, Foundation Models, Sequential Decision Making, and Embodied AI. My research develops robust and adaptive learning and reasoning methods for intelligent agents, with robotics and healthcare serving as demanding real-world validation environments.

ACADEMIC APPOINTMENTS

Purdue University, West Lafayette, IN, USA
Postdoctoral Researcher, Edwardson School of Industrial Engineering, Nov 2024 – Present
Supervisor: Dr. Juan P. Wachs

Purdue University, West Lafayette, IN, USA
Graduate Assistant (Research and Teaching), Dept. of Computer Science, Jan 2019 – Oct 2024

University of Virginia, Charlottesville, VA, USA
Graduate Assistant (Research and Teaching), Dept. of Computer Science, Aug 2015 – Dec 2018

BRAC University, Dhaka, Bangladesh
Lecturer, Dept. of Computer Science and Eng., Feb 2013 – Jul 2015

EDUCATION

Purdue University, West Lafayette, IN, USA
Doctor of Philosophy (Ph.D.), Computer Science, Oct 2024
Advisor: Yexiang Xue

University of Virginia, Charlottesville, VA, USA
Master of Science (M.S.), Computer Science, Dec 2018
Advisor: Baishakhi Ray

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
Bachelor of Science (B.Sc.), Computer Science and Engineering, Feb 2013
Advisor: Masud Hasan

SELECTED RESEARCH CONTRIBUTIONS

- **Embodied Autonomy:** Research on surgical activity recognition, cross-robot skill transfer, the DESK benchmark, and semi-autonomous teleoperation, including SARTRES and DESERTS.
- **Learning:** Robust Policy Optimization (RPO), reinforcement-learning generalization, and natural-language state representations. RPO is CleanRL's recommended default for continuous-control tasks, is integrated into `skrl`, and is used in Physical Intelligence's released implementation of real-time action chunking for vision-language-action policies.
- **Reasoning:** Multimodal medical AI and knowledge-guided vision-language reasoning, including burn assessment and clinical decision support.

- **Research Direction:** Bringing learning, procedural reasoning, and human–robot collaboration together for adaptive embodied autonomy, including surgical robotics.

RESEARCH FUNDING

Awarded Support

- **NSF Robust Intelligence Grant:** *EAGER: Theoretical Foundations for Integrating Foundational Models into Reinforcement Learning* (Award #2521982), Key Personnel, **$300,000**, 2025–2027. PI: Juan P. Wachs.
- **NAIRR Compute Support (NAIRR260368):** Principal Investigator (PI); 500 node hours on the DOE Argonne National Laboratory AI Testbed (Cerebras CS-3); September 28–December 27, 2026.
- **NAIRR Compute Support (NAIRR250333):** $25K OpenAI credits and 50K GH200 GPU hours; Lead Member; 2025–2026. PI: Juan P. Wachs.
- **Health of the Forces Pilot Funding Program, Purdue University:** *Accelerated Expertise: AI-Powered Diagnostic Pathways for Rapid Clinical Mastery of Burns*, Team Lead, **$10,000**, 2025–2026. PI: Juan P. Wachs.
- **Clinical and Translational Sciences Institute (CTSI):** *Novel Ultrasound Therapy and AI-Guided Modeling for Burn Treatment*, Team Lead, **$15,000**, 2026–2027. PI: Juan P. Wachs.

Submitted / Under Review

- **NSF TTP-Translate (NSF TTP-T)**, Co-PI, under review.
- **NSF Foundational Research in Robotics (FRR)**, Key Personnel, under review.
- **NSF AI Datasets (Planning)**, Key Personnel, under review.

Proposals in Preparation

- **NSF X-Labs, AI for Physical Systems (Phase 0)**, in preparation (2026). Selected as one of two proposals in Purdue University's institution-wide limited-submission competition.
- **NIH R21**, under preparation.

PUBLICATIONS

Peer-Reviewed Conference Papers

13. [**AAAI 2027**] *ARISE: Adaptive Agentic Reasoning with Image-grounded Self-Evaluation for Interpretable IBD Assessment.*
Pronoma Banerjee, Anuva Shah, Jason Wu, **Md Masudur Rahman**, Sanjay Mohanty, Satya Kurada, and Juan Wachs. Accepted for publication in Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2027. Innovative Applications of Artificial Intelligence (**IAAI-27**).

12. [**AAAI 2026**] *Automated Unified Reasoning with Vision-Language Models for Multi-modal Burn Assessment.* **Md Masudur Rahman**, Mohamed El Masry, Gayle Gordillo, and Juan P Wachs. In Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2026. **Oral**, Innovative Applications of Artificial Intelligence (**IAAI-26**), Acceptance rate: 15%.

11. [**MIUA 2025**] *Knowledge-Driven Hypothesis Generation for Burn Diagnosis from Ultrasound with Vision-Language Model.*
Md Masudur Rahman, Mohamed El Masry, Gayle Gordillo, and Juan P Wachs. In Proceedings of the Annual Conference on Medical Image Understanding and Analysis, 2025. **[Best Paper Award]**

10. [**NAACL 2024**] *Natural Language-based State Representation in Deep Reinforcement Learning.*
Md Masudur Rahman and Yexiang Xue. In Proceedings of the Annual Conference of the North American Chapter of the Association for Computational Linguistics, Findings, 2024.

9. [**ECML-PKDD 2022**] *Bootstrap State Representation using Style Transfer for Better Generalization in Deep Reinforcement Learning.*
Md Masudur Rahman and Yexiang Xue. In Proceedings of the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases, 2022.

8. [**ICMLA 2022**] *Bootstrap Advantage Estimation for Policy Optimization in Reinforcement Learning.*
Md Masudur Rahman and Yexiang Xue. In Proceedings of the IEEE International Conference on Machine Learning and Applications, 2022.

7. [**ICRA 2021**] *DESERTS: Delay-Tolerant Semi-Autonomous Robot Teleoperation for Surgery.*
Glebys Gonzalez, Mridul Agarwal, Mythra V. Balakuntala, **Md Masudur Rahman**, Upinder Kaur, Richard Voyles, Vaneet Aggarwal, Yexiang Xue, Juan Wachs. In Proceedings of the IEEE International Conference on Robotics and Automation, 2021.

6. [**RO-MAN 2021**] *Sequential Prediction with Logic Constraints for Surgical Robotic Activity Recognition.*
Md Masudur Rahman, Richard Voyles, Juan Wachs, Yexiang Xue. In Proceedings of the 30th IEEE International Conference on Robot & Human Interactive Communication - 2021.

5. [**RO-MAN 2021**] *Dexterous Skill Transfer between Surgical Procedures for Teleoperated Robotic Surgery.*
Mridul Agarwal, Glebys Gonzalez, Mythra V. Balakuntala, **Md Masudur Rahman**, Vaneet Aggarwal, Richard M. Voyles, Yexiang Xue, Juan Wachs. In Proceedings of the 30th IEEE International Conference on Robot & Human Interactive Communication - 2021.

4. [**RO-MAN 2019**] *Transferring Dexterous Surgical Skill Knowledge between Robots for Semi-autonomous Teleoperation.*
Md M. Rahman*, N. Sanchez-Tamayo*, G. Gonzalez, M. Agarwal, V. Aggarwal, R. M. Voyles, Y. Xue, and J. Wachs [* equal authorship], In Proceedings of the 28th IEEE International Conference on Robot and Human Interactive Communication, 2019.

3. [**IROS 2019**] *DESK: A Robotic Activity Dataset for Dexterous Surgical Skills Transfer to Medical Robots.*
N. Madapana*, **Md M. Rahman***, N. Sanchez-Tamayo*, M. V. Balakuntala, G. Gonzalez, J. P. Bindu, L. N. V. Venkatesh, X. Zhang, J. B. Noguera, T. Low, R. Voyles, Y. Xue, J. Wachs. [* equal authorship]. In Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems, 2019.

2. [**SCAM 2019**] *Toward Optimal Selection of Information Retrieval Models for Software Engineering Tasks.*
Md Masudur Rahman, Saikat Chakraborty, Gail Kaiser, and Baishakhi Ray, In 19th IEEE International Working Conference on Source Code Analysis and Manipulation.

1. [**MSR 2018**] *Evaluating How Developers Use General-Purpose Web-Search for Code Retrieval.*
Md M. Rahman, J. Barson, S. Paul, J. Kayani, F. A. Lois, S. F. Quezada, C. Parnin, K. T. Stolee, B. Ray. In Proceedings of the 15th International Conference on Mining Software Repositories, 2018.

Peer-Reviewed Journal Articles

8. [**MilMed 2026**] *A Multimodal Artificial Intelligence Reasoning Framework for Burn Diagnosis.*
Md Masudur Rahman*, Mohamed El Masry*, Gayle Gordillo, and Juan P. Wachs. *Military Medicine*, 191(Supplement_1), 598–604, 2026. *Equal contribution. DOI.

7. [**Annals of Surgery 2026**] *Automated Noninvasive Burn Diagnostic System for Health Care Using Artificial Intelligence: AMBUSH-AI.*
Mohamed El Masry*, **Md Masudur Rahman***, Surya C. Gnyawali, Thomas Smeenge, Yexiang Xue, Brett C. Hartman, Leigh Spera, Lylah H. Hutson, George E. Sandusky, Juan P. Wachs*, and Gayle Gordillo*. Annals of Surgery, 284(2), 296–304, 2026. DOI. *Equal Contribution.

6. [**MilMed 2026**] *An Artificial Intelligence Copilot for First Response Medicine.*
Yupeng Zhuo, **Md Masudur Rahman**, Eddie Zhang, Aditya Pachpande, Xiangchen Yu, Andrew W. Kirkpatrick, Kyle Couperus, Christopher Colombo, Jessica McKee, Oanh Tran, Jonah Beck, DeAnna DeVane, Chad Gorbatkin, Ross Candelore, Eleanor Birch, Juan P. Wachs. Military Medicine, 191(Supplement_1), 251–258, 2026. DOI.

5. [**JMIR 2025**] *AI-Driven Integrated System for Burn Depth Prediction With Electronic Medical Records: Algorithm Development and Validation.*
Md Masudur Rahman, Mohamed El Masry, Surya Gnyawali, Yexiang Xue, Gayle Gordillo, Juan P. Wachs. JMIR Medical Informatics, 2025.

4. [**MilMed 2025**] *A Framework for Advancing Burn Assessment With Artificial Intelligence.*
Md Masudur Rahman, Mohamed El Masry, Surya C Gnyawali, Yexiang Xue, Gayle Gordillo, Juan P Wachs. Military Medicine, 190(Supplement_2), 387–393, 2025. DOI.

3. [**CMBBE Journal 2021**] *SARTRES: a semi-autonomous robot teleoperation environment for surgery.*
Md Masudur Rahman, Mythra V. Balakuntala, Glebys Gonzalez, Mridul Agarwal, Upinder Kaur, Vishnunandan L. N. Venkatesh, Natalia Sanchez-Tamayo, Yexiang Xue, Richard M. Voyles, Vaneet Aggarwal, Juan Wachs. Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization, 9(4), 376–383, 2021. Published online November 5, 2020. DOI.

2. [**MilMed 2021**] *From the Dexterous Surgical Skill to the Battlefield—A Robotics Exploratory Study.*
Glebys T. Gonzalez*, Upinder Kaur*, **Md Masudur Rahman***, Vishnunandan Venkatesh, Natalia Sanchez, Gregory Hager, Yexiang Xue, Richard Voyles, Juan Wachs. [* equal authorship], Military Medicine, 186(Supplement_1), 288–294, 2021. DOI.

1. [**IEEE Access 2018**] *Recommending GitHub Projects for Developer Onboarding.*
Chao Liu, Dan Yang, Xiaohong Zhang, Baishakhi Ray, **Md Masudur Rahman**. In IEEE Access, vol. 6, pp. 52082-52094, 2018.

Peer-Reviewed Short Papers / Abstracts

12. [**Focused Ultrasound 2026**] *Interpretable equation discovery for optimizing splenic focused ultrasound treatment of inflammatory arthritis.*
Pronoma Banerjee, **Masudur Rahman**, Nikolaos Dervisis, Craig J. Goergen, Juan P. Wachs. Abstract accepted for poster presentation at Celebrating Focused Ultrasound: The Future is Now, November 11–13, 2026, McLean, Virginia.

11. [**MHSRS 2026**] *The CARE Center: Advancing AI-Enabled Robotics for Military and Austere Healthcare.*
Masudur Rahman, Gabriel Gruionu, Andrew D. Wisneski, David A. Peterson, Andrew A. Gonzalez, Fatih Zor, Sanjay Mohanty, Juan P. Wachs. Conference abstract, poster presentation, Military Health System Research Symposium (MHSRS), 2026.

10. [**MHSRS 2026**] *A hypothesis-driven framework for automatic gastrointestinal diagnosis from ultrasound in austere settings.*
Banerjee, P., **Rahman, M. M.**, Shah, A., Mohanty, S., Zor, F., Wachs, J. P. Conference abstract, oral presentation, Military Health System Research Symposium (MHSRS), 2026.

9. [**MHSRS 2025**] *A Chain-of-Thought AI Reasoning Framework for Burn Diagnosis.*
Md Masudur Rahman, Mohamed El Masry, Gayle Gordillo, Juan P. Wachs. Abstract Paper at Military Health System Research Symposium (MHSRS), 2025.

8. [**PSTM 2025**] *Burn Depth Diagnostic System for Healthcare Using Artificial Intelligence.*
Mohamed El Masry, **Md Masudur Rahman**, Surya Gnyawali, Juan P. Wachs, Gayle Gordillo. Abstract accepted for presentation at Plastic Surgery The Meeting (PSTM) 2025, organized by the American Society of Plastic Surgeons (ASPS).

7. [**MIUA 2025**] *A Scalable AI Copilot for High Stake Medical Scenarios.*
Yupeng Zhuo, Eddie Zhang, Aditya Pachpande, Xiangchen Yu, Krish Iyengar, Nihar Turlapati, Xiyue Chen, Trung Duc Lam, Andrew W. Kirkpatrick, Kyle Couperus, Christopher Colombo, Jessica McKee, **Md Masudur Rahman**, Juan P. Wachs. Abstract Paper at the Annual Conference on Medical Image Understanding and Analysis (MIUA), 2025.

6. [**MHSRS 2024**] *Advancing Burn Depth Assessment through Ultrasound Imaging with Vision Transformers.*
Md Masudur Rahman, Mohamed El Masry, Surya Gnyawali, Gayle Gordillo, Yexiang Xue, Juan P. Wachs. Abstract Paper at Military Health System Research Symposium (MHSRS), 2024.

5. [**MHSRS 2022**] *ASAP: A Semi-Autonomous Precise robotic framework for remote surgery under delays.*
Glebys Gonzalez, Mythra Balakuntala, Mridul Agarwal, **Md Masudur Rahman**, Thomas Low, Vaneet Aggarwal, Yexiang Xue, Richard M. Voyles, Juan Wachs. Abstract Paper at Military Health System Research Symposium (MHSRS), 2022.

4. [**MHSRS 2021**] *A Semi-autonomous Robotic Framework for Remote Surgery under Delays.*
Glebys Gonzalez, Mythra Varun Balakuntala Srinivasa Murthy, Mridul Agarwal, **Md Masudur Rahman**, Richard M. Voyles, Vaneet Aggarwal, Yexiang Xue, Juan Wachs. Abstract Paper at Military Health System Research Symposium (MHSRS), 2021.

3. [**MHSRS 2020**] *ASTRO: A Semi-Autonomous Telemedicine Robot for Operative Surgery.*
Glebys Gonzalez, **Md Masudur Rahman**, Mridul Agarwal, Mythra Balakuntala, Vishnu Venkatesh, Vaneet Aggarwal, Yexiang Xue, Richard Voyles, Gregory Hager, MAJ Andrew W Kirkpatrick, MAJ Steve Overholser, Juan Wachs. Abstract Paper at Military Health System Research Symposium (MHSRS), 2020.

2. [**ICSE '18 Companion**] *Which Similarity Metric to Use for Software Documents? A study on Information Retrieval based Software Engineering Tasks.*
Md M. Rahman, S. Chakraborty, B. Ray, In Proceedings of 40th International Conference on Software Engineering Companion, 2018.

1. [**SIGIR 2016**] *Topic Model based Privacy Protection in Personalized Web Search.*
W. Ahmad, **Md M. Rahman**, H. Wang, In Proceedings of the 39th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2016.

Peer-Reviewed Workshop Papers

8. [**ARLET@NeurIPS 2025**] *Robust Policy Gradient Optimization through Action Parameter Perturbation in Reinforcement Learning.*
Md Masudur Rahman, Juan P. Wachs, and Yexiang Xue.
Peer-reviewed, non-archival workshop paper. Aligning Reinforcement Learning Experimentalists and Theorists Workshop (ARLET) at the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025).

7. [**ARLET@NeurIPS 2025**] *Intent-Based Reward Inference for Value-Aligned Reinforcement Learning.*
Md Masudur Rahman and Juan P. Wachs.
Peer-reviewed, non-archival workshop paper. Aligning Reinforcement Learning Experimentalists and Theorists Workshop (ARLET) at the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025).

6. [**NeurIPS Workshops 2025**] *Improvisational Reasoning with Vision-Language Models for Grounded Procedural Planning.*
Md Masudur Rahman, Yupeng Zhuo, and Juan P. Wachs.
Peer-reviewed, non-archival workshop paper. Multimodal Algorithmic Reasoning Workshop (MAR), GenAI for Health: Potential, Trust, and Policy Compliance (GenAI4Health), and Embodied World Models for Decision Making (EWM) at the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025).

5. [**NeurIPS Workshops 2025**] *Vision-Language Reasoning for Burn Depth Assessment with Structured Diagnostic Hypotheses.*
Md Masudur Rahman, Mohamed El Masry, Kristo Nuutila, Gayle Gordillo, and Juan P. Wachs.
Peer-reviewed, non-archival workshop paper. GenAI for Health: Potential, Trust, and Policy Compliance (GenAI4Health), and Embodied World Models for Decision Making (EWM) at the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025).

4. [**FMDM@NeurIPS 2023**] *Natural Language-based State Representation in Deep Reinforcement Learning.*
Md Masudur Rahman and Yexiang Xue. Foundation Models for Decision Making Workshop at the thirty-seventh Annual Conference on Neural Information Processing Systems, 2023.

3. [**RRL@ICLR 2023**] *Accelerating Policy Gradient by Estimating Value Function from Prior Computation in Deep Reinforcement Learning.*
Md Masudur Rahman and Yexiang Xue. Workshop on Reincarnating Reinforcement Learning at the Eleventh International Conference on Learning Representations, 2023.

2. [**MoDeM@RLDM 2019**] *Morality in Decision-Making: A Causal Approach.*
Md M. Rahman. Accepted for oral presentation at RLDM Workshop on Moral Decision Making, 2019. [MoDeM Link][RLDM Link] [Video].

1. [**NLSE@FSE 2016**] *Finding Similar Projects in GitHub using Word2Vec and WMD.*
Md M. Rahman. Workshop on the Naturalness of Software at the 24th ACM SIGSOFT International Symposium on Foundations of Software Engineering, 2016. [Slides]

Preprints and Technical Reports

2. *Open RL Benchmark: Comprehensive Tracked Experiments for Reinforcement Learning.* [arXiv] [pdf] [GitHub] Shengyi Huang, Quentin Gallouédec, Florian Felten, Antonin Raffin, Rousslan Fernand Julien Dossa, Yanxiao Zhao, Ryan Sullivan, Viktor Makoviychuk, Denys Makoviichuk, Mohamad H. Danesh, Cyril Roumégous, Jiayi Weng, Chufan Chen, **Md Masudur Rahman**, João G. M. Araújo, Guorui Quan, Daniel Tan, Timo Klein, Rujikorn Charakorn, Mark Towers, Yann Berthelot, Kinal Mehta, Dipam Chakraborty, Arjun KG, Valentin Charraut, Chang Ye, Zichen Liu, Lucas N. Alegre, Alexander Nikulin, Xiao Hu, Tianlin Liu, Jongwook Choi, Brent Yi. 2024

1. *A Case Study on the Impact of Similarity Measure on Information Retrieval based Software Engineering Tasks.* **Md Masudur Rahman**, Saikat Chakraborty, Gail Kaiser, Baishakhi Ray. Technical Report 2018.

RESEARCH SOFTWARE, BENCHMARKS, AND DATASETS

- **Robust Policy Optimization (RPO):** Open implementation; recommended default for continuous-control tasks in CleanRL; integrated into `skrl`; used in Physical Intelligence's released implementation of real-time action chunking for vision-language-action policies.
- **Open RL Benchmark:** Contributor to a tracked experimental benchmark for reinforcement-learning evaluation and reproducibility.
- **DESK:** Co-first author; co-led the robotic system work with a focus on AI/ML for dexterous surgical skill transfer across medical robots.

AWARDS & HONORS

- **Gold Reviewer**, ICML, 2026.
- **Best Paper Award**, Medical Image Understanding and Analysis (MIUA), 2025.
- **Outstanding Reviewer, Top 10%**, ICML, 2022.
- Postdoctoral Travel Award, Purdue University, Fall 2025.
- Postdoctoral Travel Award, Purdue University, Winter 2025.
- GenAI4Health Travel Grant, NeurIPS 2025.

TEACHING EXPERIENCE

Instructor

BRAC University, Dhaka, Bangladesh

- CSE 220: Data Structure (Undergraduate), Fall 2014, Spring 2015, Summer 2015.
- CSE 260: Digital Logic Design (Undergraduate), Summer 2015.
- CSE 101: Introduction to Computer (Undergraduate), Fall 2014, Spring 2015.

Co-Instructor / Guest Lectures

Purdue University, West Lafayette, IN, USA

- IE 549: Machine Vision in Intelligent Robotic Systems, 2025. Role: Guest lectures.

Purdue University, West Lafayette, IN, USA
Co-teaching a course on AI in medicine and large language models.

Teaching Assistant

Purdue University, West Lafayette, IN, USA

- CS 578: Statistical Machine Learning (Graduate), Spring 2021.
- CS 47300: Web Information Search and Management (Undergraduate), Fall 2020.

University of Virginia, Charlottesville, VA, USA

- CS 4780: Introduction to Information Retrieval (Graduate), Fall 2018.
- CS 6501: Data Science for Software Engineering (Graduate), Spring 2016.
- CS 3102: Theory of Computation (Undergraduate), Fall 2015.
- CS 3330: Computer Architecture (Undergraduate), Fall 2015, Spring 2016, Fall 2018.

STUDENT RESEARCH MENTORING

Current Graduate Researchers

Yupeng Zhuo (2024–Present)
Ph.D. Student, Edwardson School of Industrial Engineering, Purdue University
Topic: AI-Copilot for Emergency Medicine

Pronoma Banerjee (2025–Present)
Ph.D. Student, Edwardson School of Industrial Engineering, Purdue University
Topic: Reinforcement Learning with Foundation Model (LLM/VLM)

Maxwell Ryan Kawada (2025–Present)
Ph.D. Student, Biomedical Engineering, Purdue University
Topic: Medical Robotics

Previous Student Researchers

Zachery Peter Berg, Purdue University, 2021–2022
B.Sc. in Computer Science, 2021; M.S. in Computer Science, 2022
Topic: Double Descent in Deep Reinforcement Learning

Peter Khlebnikov, Purdue University, 2022
B.Sc. in Computer Science, 2022
Topic: Exploration in Deep Reinforcement Learning

Brian Yifei Sun, Purdue University, 2021
B.Sc. in Computer Science, 2021
Topic: Reinforcement Learning

Chao Liu, Chongqing University, China, 2018
Ph.D. in Software Engineering, 2018
Topic: Recommending GitHub Project for Developer Onboarding
Paper: IEEE Access 2018
Appointment (2023): Associate Professor, Chongqing University, China

Jed Barson, University of Virginia, USA, 2018
B.Sc. in Computer Science, 2018
Topic: Code Search; Paper: MSR 2018
First Job Appointment: Software Engineer at Cisco

Eliza Yixuan Nie, University of Virginia, USA, 2017
B.Sc. in Computer Science, 2017
Topic: GitHub Project Search
First Job Appointment: Software Engineer at Facebook

INVITED TALKS

- Guest Lecture, CS-4390/5390: Artificial Intelligence for Scientific Discovery, The University of Texas at El Paso, 2025. Topic: AI for High-stakes Environment (Healthcare).

PROFESSIONAL SERVICE

Federal Review Panels

- Reviewer Panel, National Artificial Intelligence Research Resource (NAIRR) Pilot, October 2025–Present.

Conference Organization

- Organizer and Chair, Birds of a Feather session on Vision-Language Models in Medical Surgery, NAACL 2024.

Program Committee Service

- Senior Program Committee member, AAAI 2027, AI for Social Impact (AISI) track.

Conference Reviewing

- NAACL: 2027 (ARR October cycle).
- HRI: 2027.
- ICML: 2022, 2023, 2024, 2025, 2026.
- NeurIPS: 2022, 2023, 2025.
- ICLR: 2024, 2026, 2027.
- AAAI: 2026 (Main Track, AI Alignment Track, AI for Social Good Track).
- AISTATS: 2023, 2026; UAI: 2023, 2024; AAMAS: 2022, 2023, 2024.
- CVPR: 2026; WACV: 2026; BMVC: 2025; MICCAI: 2025; MIUA: 2025.
- ECML-PKDD: 2021; ICRA: 2021; ICMLA: 2021; RO-MAN: 2021.

Other Peer-Review Service

- Multimodal Algorithmic Reasoning (MAR@NeurIPS) Workshop, 2025.
- Aligning Reinforcement Learning Experimentalists and Theorists (ARLET@NeurIPS) Workshop, 2025.
- Sub-reviewer: ICLR (2023), NeurIPS (2019, 2020, 2021), ICML (2019, 2021), IJCAI (2019, 2021), UAI (2020, 2021), AISTATS (2020), AAMAS (2021), KDD-DMAIC Workshop (2019), VLDB Journal (2019).

Journal Reviewing

- DIGITAL HEALTH (Sage), 2026.
- Nature – Scientific Data, 2026.
- IEEE Robotics and Automation Letters (RA-L), 2023, 2024, 2025.
- MHSRS/Military Medicine, 2022–2025.
- IEEE Access, 2018.

Institutional Service

- Faculty Search Committee Representative, Computer Science Graduate Student Association (CSGSA), Purdue University, 2022–2023.
- General Secretary, Bangladesh Students Association (BDSA), Purdue University, 2019–2020.

Last update: November 4, 2026